

Final Exam

Name: _____

Please circle your section number!

SECTION 1	SECTION 2	SECTION 3	SECTION 4
AMY COCHRAN (MWF 9:55p)	BENEDEK VALKÓ (MWF 12:05p)	SARAH TAMMEN (TuTh 9:30a)	SHAOMING GUO (MWF 8:50a)

- **There are 6 problems on the exam, some of them have multiple parts.** The problems are on pages 3 to 14. Make sure to read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.** You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Your answer can be left in terms of the cumulative distribution function Φ of a standard normal random variable or in terms of the exponential function e^x , e.g. expressions like $\Phi(1)$ or $3e^{-9}$ would not need to be simplified further.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	20	
2	15	
3	20	
4	19	
5	10	
6	16	
Total:	100	

List of named distributions

$$X \sim \text{Bernoulli}(p), \quad \text{PMF: } p_X(0) = 1 - p, \quad p_X(1) = p$$

$$X \sim \text{Binomial}(n, p), \quad \text{PMF: } p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad 0 \leq k \leq n$$

$$X \sim \text{Geometric}(p), \quad \text{PMF: } p_X(k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

$$X \sim \text{Poisson}(\lambda), \quad \text{PMF: } p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

$$X \sim \text{Hypergeometric}(N, N_A, n), \quad \text{PMF: } p_X(k) = \frac{\binom{N_A}{k} \binom{N - N_A}{n - k}}{\binom{N}{n}}, \quad 0 \leq k \leq n$$

$$X \sim \text{Uniform}(a, b), \quad \text{PDF: } f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Normal}(\mu, \sigma^2), \quad \text{PDF: } f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}, \quad t \in \mathbb{R}$$

$$X \sim \text{Exponential}(\lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

$$X \sim \text{Gamma}(r, \lambda), \quad \text{PDF: } f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

Here $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$, and $\Gamma(n) = (n-1)!$ for $n \geq 1$ integer.

$$(X_1, \dots, X_k) \sim \text{Multinomial}(n, k, p_1, \dots, p_k),$$

$$\text{joint PMF: } p(a_1, \dots, a_k) = \frac{n!}{a_1! a_2! \dots a_k!} p_1^{a_1} \dots p_k^{a_k} \text{ for } a_1, \dots, a_k \geq 0 \text{ with } a_1 + \dots + a_k = n.$$

1. For each of the following problems, we suppose that $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) random variables with mean 0 and variance 1.

(a) (5 points) What is the expected value of the random variable $Y = X_1 + 2X_2 + 3X_3X_4$?

(b) (5 points) Find $\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3 + 1)$.

(Problem 1 continued)

- (c) (5 points) Suppose that we know that the moment generating function of X_1 is given by $M(t) = \frac{1}{1-t^2/2}$ whenever $|t| < \sqrt{2}$. What is the moment generating function of the random variable $Z = X_1 + X_2$ at $t = 1$?

- (d) (5 points) Suppose that X_1 has standard normal distribution. Find $P(W > 0)$ for $W = X_1 + 2X_2 - X_3$.

2. Jrue is a professional basketball player, averaging 16 points per game. Let X denote the number of points he scores in his next basketball game.

Note. You will estimate and/or approximate certain probabilities in some of the following problems. Justify the approach you use!

- (a) (5 points) What is the best upper bound you can provide for $P(X \geq 30)$?

- (b) (5 points) Suppose that we also learn that the variance of X is approximately 36. How can you improve on your upper bound on $P(X \geq 30)$?

(Problem 2 continued)

- (c) (5 points) Approximate the probability that in the next season (100 games, counting playoffs), Jrue will score more than 1700 points in total.

Note. You may assume that the number of points scored by Jrue in different games are independent and identically distributed.

3. We flip a fair coin. If it is heads, we choose a number uniformly at random from $[0, 1]$. If it is tails, we choose a number uniformly at random from $[0, 2]$. Let X denote the chosen number.

(a) (5 points) Find the probability that $X \leq 1/2$.

(b) (5 points) Find the probability that we rolled heads given that $X \leq 1/2$.

(Problem 3 continued)

- (c) (5 points) Find the cumulative distribution function of X . (Make sure that the function you provide is defined on the whole line.)

(Problem 3 continued)

(d) (5 points) Show that X is continuous, and find its probability density function.

4. Suppose we have jointly continuous X and Y with joint probability density function:

$$f(x, y) = \begin{cases} \frac{4}{y}e^{-4y} & \text{if } 0 < x < y, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) (6 points) Let $f_Y(y)$ denote the probability density function of Y . Find $f_Y(y)$ and identify the distribution of Y by name.

Note. Make sure to identify the parameters of the distribution.

- (b) (6 points) Let $f_{X|Y}(x|y)$ denote the conditional density function of X given $Y = y$. Find $f_{X|Y}(x|y)$ and identify the conditional distribution of X given $Y = y$ by name.

Note. Make sure to identify the parameters of the distribution.

(Problem 4 continued)

(c) (3 points) Find $E[X|Y]$.

(d) (4 points) Find $E[X]$.

5. Suppose that X and Y are independent continuous random variables with probability density functions f_X and f_Y .

(a) (3 points) Write down the integral formula for the probability density function of $Z = X + Y$.

(b) (7 points) Suppose that $X \sim N(0, 1)$ and $Y \sim \text{Unif}[0, 1]$. Find the probability density function of $Z = X + Y$ at $z = 7$ using the integral formula given in (a).

Note. Your final answer should be in terms of the Φ function.

6. A monkey is sitting in front of a keyboard that has the 26 capital letters of the alphabet (and nothing else). He starts randomly hitting the keys: he chooses the keys independently and uniformly.
- (a) (4 points) What is the probability that the first two keystrokes are U and W in this order?
- (b) (8 points) We print out the first 1000 keystrokes (in order). Let X be the number of times we see the sequence UW spelled out (these 2 letters next to each other) somewhere in that text. Find $E[X]$.

(Problem 6 continued)

(c) (4 points) Find $\text{Var}(X)$ for the random variable X in part (b).

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